

Micro I, class 6

Starting from the identity $d(p, e(p, \bar{u})) = h(p, \bar{u})$ (Lemma 1 in my notes), and under the assumption that the expenditure function $e(\cdot, \cdot)$ is C^2 with $\partial e / \partial \bar{u} > 0$, $h(\cdot, \bar{u})$ and d are differentiable and

$$\frac{\partial h_i}{\partial p_j} = \underbrace{\frac{\partial d_i}{\partial p_j} + d_j \frac{\partial d_i}{\partial w}}_{s_{ij}}. \quad (1)$$

This equality and Corollary 1 from my notes on *Demand Properties* give us testable implications of the ordinary demand: d satisfies budget balance; it is homogenous of degree 0; and the matrix whose i, j element, is s_{ij} – the right side of (1) – is a *symmetric, negative semidefinite* matrix. In particular $s_{ii}(p, w) \leq 0$ and $s_{ij}(p, w) = s_{ji}(p, w)$ for $i = 1, \dots, L$ and every $(p, w) \gg 0$.

Rearrange (1) to produce the *Slutsky equation*, decomposing the (total) effect of price changes (TE) into substitution (SE) and wealth (WE) effects,

$$\underbrace{\frac{\partial d_i}{\partial p_j}}_{\text{TE}} = \underbrace{\frac{\partial h_i}{\partial p_j}}_{\text{SE}} - \underbrace{d_j \frac{\partial d_i}{\partial w}}_{\text{WE}}.$$

We then discussed another approach to consumer theory, *revealed preference*, developed by Samuelson (1938). I introduced a revealed preference relation, $\mathcal{R}$, and the *weak axiom of revealed preference* (WARP). Applied to a function $f : \mathbb{R}_{++}^{L+1} \rightarrow \mathbb{R}_+^L$ that satisfies budget balance— $p \cdot f(p, w) = w$ for every $(p, w) \gg 0$ —the function f satisfies WARP if, for any positive price-wealth pairs (p, w) and (p', w')

$$[p' \cdot f(p, w) \leq w' \text{ and } f(p, w) \neq f(p', w')] \implies p \cdot f(p', w') > w.$$

We noted that if a preference representation is continuous, LNS, and *strictly quasiconcave*, then the resulting consumer demand d from the (UM) problem is single-valued, satisfies budget balance, and satisfies WARP. We will use this fact to compare two taxes: a vector $t \in \mathbb{R}^L$ of per-unit taxes; and a lump-sum tax of T , adjusted so that the two taxes are revenue equivalent. We asked two questions: which tax, if any, is better; and how do the *choices* under the two taxes compare? On Thursday we will discuss the answers to these questions. The second will lead us to the *compensated law of demand* (CLD, eq 2.F.1 in MWG). It asserts that, for any $(p, w) \gg 0$, any $\Delta p \in \mathbb{R}^L$ with $p + \Delta p \gg 0$

$$\Delta p \cdot [f(p + \Delta p, w + \Delta w) - f(p, w)] \leq 0,$$

where $\Delta w = \Delta p \cdot f(p, w)$. The compensation Δw just allows the consumer to buy $f(p, w)$ at prices $p + \Delta p$. It is sometimes called a *Slutsky compensation*, to distinguish it from a *Hicks compensation*, which adjusts wealth to keep utility constant.